

JOSH JACOBSON

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EDUCATION

Ph.D. in Statistics

Aug 2020–Dec 2025

University of Wollongong, Wollongong, NSW

Dissertation*: *Statistical Methods for Spatio-Temporal Prediction of CO₂ Flux from Multiprocess Satellite Data*

Advisors: Noel Cressie, Andrew Zammit Mangion, and Michael Bertolacci

*Examiners' Commendation for Outstanding Dissertation

M.S. in Applied Mathematics

Aug 2018–May 2020

University of Colorado, Boulder, CO

Thesis: *Beyond Univariate Calibration: Verifying Spatial Structure in Ensembles of Forecast Fields*

Advisors: William Kleiber and Michael Scheuerer

B.S. in Applied Mathematics (with honors)

Aug 2015–May 2019

University of Colorado, Boulder, CO

Minors in Computer Science, Atmospheric and Oceanic Sciences

PROFESSIONAL EXPERIENCE

Visiting Assistant Professor

Jan 2026–Now

Department of Applied Mathematics, University of Colorado, Boulder, CO

Research: physics-informed neural emulators, probabilistic forecasting

Data Science Consultant

Sep 2019–Jan 2023*

Jupiter Intelligence, Boulder, CO

Research: compound weather hazards, copula models, approximate Bayesian computation

Supervisors: Steve Sain and Alexis Hoffman

*Combination of part-time and full-time contracts

PUBLICATIONS

Google Scholar: https://scholar.google.com/citations?user=MIK_mtoAAAAJ

- [6] **Jacobson, J.**, Bertolacci, M., Zammit-Mangion, A., Schuh, A., & Cressie, N. (2025). WOMBAT v2.S: A Bayesian inversion framework for attributing global CO₂ flux components from multiprocess data. *Environmetrics*, 36(8), e70052. <https://doi.org/10.1002/env.70052>
 - Winner of 2025 EnviBayes Student Paper Competition
 - Winner of 2025 Tjanpi Award for Best Student Paper in Environmental Statistics
- [5] Cressie, N., Zammit-Mangion, A., **Jacobson, J.**, & Bertolacci, M. (2023). Earth's CO₂ battle: A view from space. *Significance*, 20(1), 14-19. <https://doi.org/10.1093/jrssig/qmad003>
- [4] **Jacobson, J.**, Cressie, N., & Zammit-Mangion, A. (2023). Spatial statistical prediction of solar-induced chlorophyll fluorescence (SIF) from multivariate OCO-2 data. *Remote Sensing*, 15(16), 4038. <https://doi.org/10.3390/rs15164038>
- [3] Vu, Q., Cao, Y., **Jacobson, J.**, Pearse, A. R., & Zammit-Mangion, A. (2021). Discussion on "Competition on Spatial Statistics for Large Datasets." *Journal of Agricultural, Biological and Environmental Statistics*, 26, 614-618. <https://doi.org/10.1007/s13253-021-00464-0>

- [2] **Jacobson, J.**, Kleiber, W., Scheuerer, M., & Bellier, J. (2020). Beyond univariate calibration: Verifying spatial structure in ensembles of forecast fields. *Nonlinear Processes in Geophysics*, 27(3), 411-427. <https://doi.org/10.5194/npg-27-411-2020>
 • NPG Paper of the Month Award, August 2020
- [1] Raseman, W. J., **Jacobson, J.**, & Kasprzyk, J. R. (2019). Parasol: An open source, interactive parallel coordinates library for multi-objective decision making. *Environmental Modelling & Software*, 116, 153-163. <https://doi.org/10.1016/j.envsoft.2019.03.005>

SELECTED AWARDS & FELLOWSHIPS

2025 **Winner, Student Paper Award**, Environmental Statistics Section of the Statistical Society of Australia
 2025 **Winner, Student Paper Competition**, EnviBayes Section of the International Society for Bayesian Analysis
 2025 **Winner, J.B. Douglas Postgraduate Awards**, Statistical Society of Australia, NSW Branch
 2025 **Best Lightning Talk**, The Bayesian Young Statisticians Meeting
 2024 **Best Student Presentation**, 31st Conference of The International Environmetrics Society
 2021 **Allison Harcourt Best Poster Award**, Early Career & Student Statisticians Conference
 2021 **Statistical Data Science Fellowship**, Australian Mathematical Sciences Institute
 2020 **Graduate Fellowship**, University of Wollongong (2020–2024)
 2020 **Paper of the Month Award**: August 2020, *Nonlinear Processes in Geophysics*

GRANTS

National Aeronautics and Space Administration, 23-OCOST23-0001 2023–2025
Role: Student Investigator Award Amount: \$0*
Project Title: Hierarchical Spatio-Temporal Statistical Methods for Analyzing OCO-2/3 Data
Funding Program: Research Opportunities in Space and Earth Science (ROSES)
 *International organizations are not eligible for funding, but the award grants membership of the OCO Science Team

SOFTWARE & DATASETS

WOMBAT v2.S: Workflow to attribute global CO₂ flux components from multiprocess satellite data
<https://github.com/joshhjacobson/wombat-v2s>

coSIF: Dataset of monthly, 0.05-degree cokriging predictions and standard errors for SIF over North America
<https://doi.org/10.5281/zenodo.8078592>

FTE: Workflow to verify spatial structure in ensemble forecasts, applied to downscaled GEFS reforecast product
<https://github.com/joshhjacobson/FTE>

Parasol: Parallel coordinates visualization library for multi-objective decision-making
<https://github.com/ParasolJS/parasol-es>

TEACHING EXPERIENCE

Instructor of Record
 University of Colorado Boulder
 Applied Regression (STAT 3400) Spring 2026

University of Wollongong	
Fundamentals of Biostatistics (STAT 251)	<i>Fall 2025</i>
Introduction to Statistics (STAT 101)	<i>Fall 2025</i>

Guest Lecturer

University of Colorado Boulder	
Analysis of Climate and Weather Observations (GEOG 3301)	<i>Mar 2026</i>

Teaching Assistant

University of Wollongong	
Mathematics for Computing (MATH 255)	<i>Spring 2025</i>
Stochastic Processes and Time Series Analysis (STAT 304/804)	<i>Fall 2024</i>
Generalized Linear Models (STAT 332/832)	<i>Spring 2023, 2024</i>
Statistical Methods for Data Science (STAT 301/801)	<i>Fall 2023</i>
University of Colorado Boulder	
Fourier Series and Boundary Value Problems (APPM 4350)	<i>Fall 2018</i>
Introduction to Programming for Engineers (CSCI 1320)	<i>Spring 2016</i>

SELECTED PRESENTATIONS

Invited Presentations

2026-06 Tjanpi Award Seminar, Environmental Statistics Section of the Statistical Society of Australia, Online

2026-04 MathGeo Seminar, Department of Applied Mathematics, University of Colorado, Boulder, CO

2026-02 NASA Orbiting Carbon Observatory-2/3 Science Team Telecon, Online

2025-11 J.B. Douglas Postgraduate Awards, Sydney, NSW

2025-11 Computational Statistics in Data Science Workshop, Wollongong, NSW

2025-09 NASA Orbiting Carbon Observatory-2/3 Science Team Meeting, Fort Collins, CO (via Teams)

2025-03 Seminar, Department of Statistics, University of New South Wales, Sydney, NSW

2023-04 Seminar, National Institute for Applied Statistics Research Australia (NIASRA), Wollongong, NSW

Contributed Presentations

2025-12 Atmospheric Composition & Chemistry Observations & Modelling Conference, Wollongong, NSW

2025-04 The Bayesian Young Statisticians Meeting (BAYSM), Online

2024-12 31st Conference of The International Environmetrics Society (TIES), Adelaide, SA

2023-12 Australian Statistical Conference, Wollongong, NSW

2023-10 NASA Orbiting Carbon Observatory-2/3 Science Team Meeting: SIF Breakout, Boulder, CO (via Teams)

2023-01 American Meteorological Society 103rd Annual Meeting, Denver, CO

2021-07 Australian and New Zealand Statistical Conference, Online

Contributed Posters

2024-10 ENVR Workshop on Spatial Data Science for the Environment, Boulder, CO

2021-07 Early Career & Student Statisticians Conference, Online

PROFESSIONAL SERVICE & LEADERSHIP

Peer Reviewer

2024–Now

Global Biogeochemical Cycles

Journal of Agricultural, Biological, and Environmental Statistics (JABES)

Mathematical Geosciences

Outreach Volunteer

Spring & Fall 2024

School of Mathematics and Applied Statistics, University of Wollongong

Head of Postgraduate Seminar Series

Spring 2024

School of Mathematics and Applied Statistics, University of Wollongong

Co-host of “Probably Novel Radio Show and Podcast”

Spring 2019

Radio 1190, Department of Applied Mathematics, University of Colorado Boulder

PROFESSIONAL DEVELOPMENT

Winter School on Statistical Data Science, Australian Mathematical Sciences Institute (AMSI)

Jul 2021

Two-week course on Bayesian statistics, likelihood-free inference, and neural networks.

PROFESSIONAL MEMBERSHIPS

American Geophysical Union

American Meteorological Society

American Statistical Association

International Society for Bayesian Analysis

The International Environmetrics Society